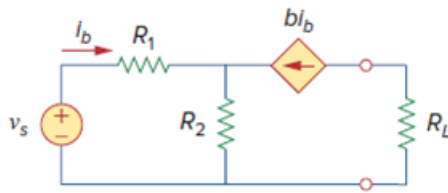


Problem

The network in Fig. 4.124 models a bipolar transistor common-emitter amplifier connected to a load. Find the Thevenin resistance seen by the load.

Figure. 4.124:



Step-by-step solution

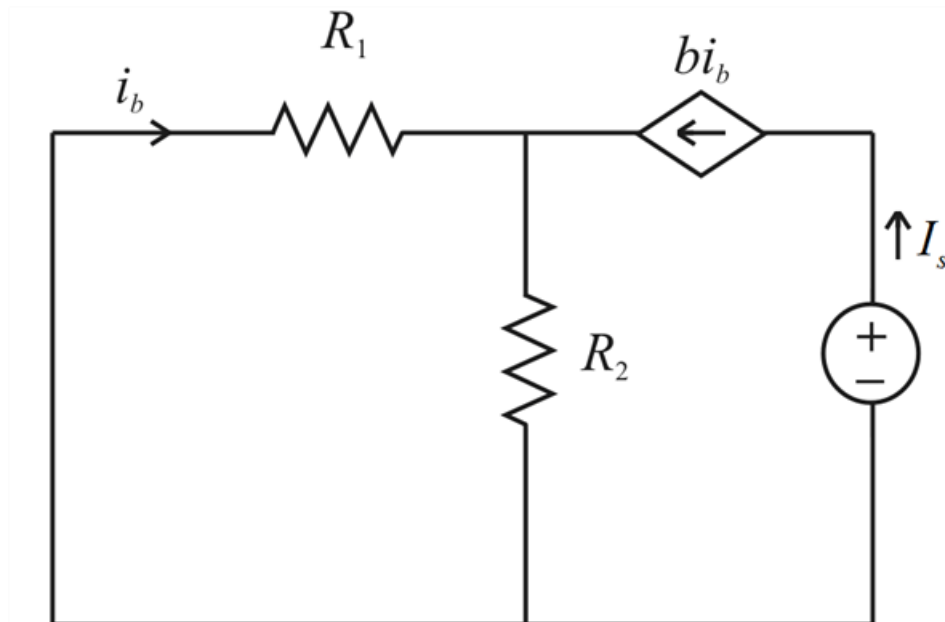
Step 1 of 3

For the given circuit we can't directly find the thevenin voltage because of the presence of dependent source. So we keep the test voltage source across the load terminals so the thevenin resistance becomes

$$R_{th} = \frac{1V}{I_s}$$

$$R_{th} = \frac{1}{I_s}$$

Step 2 of 3



Step 3 of 3

By applying KVL to the first loop

$$i_b R_1 + (i_b + b i_b) R_2 = 0$$

$$i_b (R_1 + R_2 + b R_2) = 0$$

$$(R_1 + R_2 + b R_2) \neq 0$$

$$\therefore i_b = 0$$

From the above circuit

$$I_s = b I_b$$

$$\therefore I_s = 0 \text{ A} \quad (\because i_b = 0)$$

$$R_{th} = \frac{1}{I_s}$$

$$\therefore R_{th} = \frac{1}{0}$$

$$\boxed{R_{th} = \infty}$$